

DS	Session	Hashing	Question Information	Level 1	Challenge 95
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Problem Description

You are given a string which comprises of lower case alphabets (a-z), upper case alphabets (A-Z), numbers, (0-9) and special characters like !,.,; etc.

You are supposed to find out **which character occurs the maximum number of times and the number of its occurrence**, in the given string. If two characters occur equal number of times, you have to output the character with the lower **ASCII value**.

For example, if your string was: **aaaaAAAA**, your output would be: **A 4**, because **A has lower ASCII value than a**.

Input format:

The input will contain a string.

Output format:

You've to output two things which will be separated by a space:

- i) The character which occurs the maximum number of times.
- ii) The number of its occurrence.

Constraints:

The maximum length of the string can be **1000**.

Logical Test Cases

Test Case 1

INPUT (STDIN)

puppy is a dog!!!!!!!!!!!!

EXPECTED OUTPUT

! 12

Test Case 2

INPUT (STDIN)

leana is a singer!!!!!!!!

EXPECTED OUTPUT

! 7

▼ Mandatory Test Cases

Test Case 1

KEYWORD

```
for(i=0;i<l;i++)
```

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

6

Test Case 2

TOKEN COUNT

210

Test Case 3

NLOC

35

Hashing5.c +

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main() {
5     char s[1005];
6     if (!fgets(s, sizeof(s), stdin)) return 0;
7
8     int l = strlen(s);
9     while (l > 0 && (s[l - 1] == '\n' || s[l - 1] == '\r')) {
10         s[--l] = '\0';
11     }
12
13     int freq[256] = {0};
14     int i;
15     for(i=0;i<l;i++) {
16         freq[(unsigned char)s[i]]++;
```

main | CH.SC.U4CSE25225 | Hashing5.c

Ln 32, Col 1 Spaces: 4 UTF-8 CRLF C

```
18
19     int max_cnt = -1;
20     char best_char = 0;
21
22     for (i = 0; i < 256; i++) {
23         if (freq[i] > max_cnt) {
24             max_cnt = freq[i];
25             best_char = (char)i;
```

```
26     }  
27 }  
28  
29 printf("%c %d\n", best_char, max_cnt);
```

```
30     return 0;  
31}  
32
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
puppy is a dog!!!!!!!!!!!!!!
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Hashing5.exe
! 12
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
leana is a singer!!!!!!
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Hashing5.exe
! 7
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```